Data Translation and Mapping in Spark

In this train, we will utilise Apache Spark to perform data translation and mapping, transforming raw data into structured formats using Spark's robust framework.

Learning objectives

By the end of this train you should be able to:

- [Placeholder for objective 1]
- [Placeholder for objective 2]
- [Placeholder for objective 3]

Learning Objective

- Be able to translate and map data using Spark.

Translation and Mapping

Translation is the process of making the data more human-understandable, for example, changing work order codes in a field into human-readable descriptions of the work done. This process is important to ensure the data is understandable for a customer-facing product, but can also assist data scientists and analysts in the process of creating intelligence and analytics.

Mapping is a similar process whereby we create a map between the source system and destination system, typically at the field level, for example, changing column names to conform to the destination system. This is especially relevant between different data systems (for example, JSON or XML to a relational database), where different constraints may exist on the field names.

Mapping of datasets can be done through metadata, which explains the data fields and attributes and constitutes the data and rules that govern how the data is stored

within the database or data repository.

For this particular dataset, the abbreviations of the fields above are:

- `rain` - Rain (mm)
- `temp` - Temperature (°C)
- `wetb` - Wet Bulb Air Temperature (°C)
- `dewpt` - Dew Point (°C)
- `vappr` - Vapour Pressure (hPa)
- `rhum` - Relative Humidity (%)
- `msl` - Mean Sea Level Pressure (hPa)
- `wdsp` - Mean Hourly Wind Speed (kt)
- `wddir` - Predominant Hourly Wind Direction (degrees)
- `sun` - Sun (hours)
- `vis` - Visibility (m)
- `clht` - Cloud Ceiling Height
- `clamt` - Cloud Amount (Oktas)

These field definitions will come in handy during data integrity checks, helping to validate typecasting and ensure accuracy. It may also help us confirm that we have done the typecasting correctly. We got this specifically from the metadata file that was included alongside the dataset.

Translation Vs Mapping

Translation	Mapping
Changes data values (e.g., codes to readable labels)	Changes schema (e.g., column names)
Works at the data value level	Works at the structure/field level
Example: 'WD01' → 'Welding'	Example: 'wdsp' → 'wind_speed'

Conclusion

Translation and mapping are essential steps in preparing data for both human interpretation and system integration. Translation helps make data more understandable by converting cryptic codes or abbreviations into clear, human-readable descriptions, which is valuable for end users and analysts alike. Mapping ensures consistency and compatibility between different data systems by aligning field

names and structures, often guided by metadata that defines the meaning and rules of the data. Together, these processes improve data clarity, usability, and integrity, especially when working across diverse platforms or preparing data for analysis.

▼Key Takeaways ✖

- **Translation** makes data more accessible by converting technical codes into human-readable descriptions, which improves usability for non-technical users.
- **Mapping** aligns source and destination systems by standardizing field names and structures, especially important when integrating across different data formats.
- **Metadata** plays a critical role in both translation and mapping by providing essential context, rules, and definitions for interpreting and validating the data.

▼Key Takeaways ✖

- **Takeaway 1:** [Placeholder for key takeaway 1]
- **Takeaway 2:** [Placeholder for key takeaway 2]
- **Takeaway 3:** [Placeholder for key takeaway 3]